

2027 MSA Strategic Initiatives — CALIBER Proposal (Draft)

Working draft for a proposal to the [MSA Strategic Initiatives program](#), 2027 call. Bracketed [TODO] items need lab input before submission. Citations are drawn from the Edison research reports in [outputs/msa_2027_strategic_initiatives/](#).

Call constraints (2027 cycle)

- **Deadline:** 11:59 PM Eastern Time, November 1, 2026
 - **Funding:** suggested level up to \$15,000 for a one-year project or \$20,000 total over two years; significantly higher possible for exceptional projects with major benefits to the Society; 3–4 awards expected
 - **Eligibility:** primary applicant must be a current MSA member; detailed budget required
 - **Scope:** bold, innovative *new* programs; not intended to sustain existing programs or the primary research/educational missions of institutions
 - **Primary areas of emphasis targeted** (of the six listed in the call):
 - *Area 2:* MSA-branded development and dissemination of information and open source tools related to advances in microscopy, microanalysis, data management, and image analysis
 - *Area 6:* Collection, preservation, management, and dissemination of microscopy-related information, including technical procedures, basic theories, scientific backgrounds, MSA archived material, and related resources
 - **Secondary areas touched:** student involvement (Area 1), member benefits (Area 3), academic/governmental/industrial partnerships (Area 5)
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Title

The MSA Microanalysis Parameter Commons: MSA-branded open-source tools and a preserved knowledge base for selecting high-fidelity acquisition parameters, built on CALIBER

[Alternate: "CALIBER for the Community: An MSA-Branded Open Platform and Knowledge Commons for Microanalysis Acquisition Parameters"]

Applicant team

- **Principal applicant:** [TODO: name — must be a current MSA member; confirm membership status]
- **Co-applicants / advisors:** [TODO: draw from the reviewer/collaborator shortlist — see issue 10]
- **Host lab:** Vertical Cloud Lab [TODO: institution details]

Requested support

\$20,000 over two years (*scale to \$15,000 / one year by dropping Year 2 items if preferred*)

Summary

Quantitative microscopy lives or dies by acquisition parameters. Physical standards can hold SEM-EDS to roughly $\pm 5\%$ relative uncertainty while standardless analysis can err by $\pm 30\%$, with normalization hiding the damage [1]; raising EBSD collection speed from 54 to 154 patterns per second dropped correct indexing from 34.9% to 9.3% [2]; and supplying a 12-phase candidate list instead of the correct two caused *zero* patterns to index correctly despite ideal beam settings [3]. Our own LPBF AlSi10Mg work mirrors this: standardless EDS over-reported trace Mg several-fold, and indexing yield swung from 40% to 80% on beam-current and phase-list corrections alone. The rationale behind "recommended" settings — why standards demand overvoltage ≥ 1.8 , where defaults break down — is scattered across decades of literature, standards whose derivations are rarely documented, vendor-proprietary software, and the memory of senior microscopists; when they retire, it leaves with them [1,15].

We propose an MSA-branded **Microanalysis Parameter Commons**: (1) an open-source release of **CALIBER**, a retrieval-augmented, uncertainty-aware recommender that suggests literature- and physics-grounded SEM, EDS, EBSD, and XRF acquisition parameters for a stated sample and analysis goal, and (2) a curated, versioned, citable knowledge base that collects and preserves *why* parameters are chosen — technical procedures, underlying physics, and the derivations behind standard settings — cross-linked to MSA's own archives, where much of this knowledge already sits unorganized [12,13]. An open benchmark campaign on novel alloys fills a documented evidence gap (no controlled factorial acquisition studies exist for LPBF AlSi10Mg [1,3]), students earn stipends curating entries under expert review, and results flow to members and the wider imaging community through M&M tutorials, open webinars, a Microscopy Today article, and MSA's web presence. The award funds a community resource, not our lab's research program — one scoped deliberately to MSA's mission of fostering research, innovation, advancement, and promotion of microscopy.

Alignment with the MSA mission

MSA's stated mission is to be "a collaborative community dedicated to fostering research, innovation, advancement, and promotion of microscopy" ([Mission and Vision](#)). The initiative is built to serve all four commitments, not only the two call areas it targets:

- **Fostering research** — the open factorial benchmark (O3) creates validation data the literature demonstrably lacks [1,3], and every lab that adopts validated parameters gets more trustworthy measurements (roughly $\pm 5\%$ standards-based vs. $\pm 30\%$ standardless relative uncertainty in SEM-EDS [1]).

- **Innovation** — a retrieval-augmented, uncertainty-aware acquisition recommender (O1) is a new kind of open microscopy tool: connective infrastructure over HyperSpy, kikuchipy, and the rest of an ecosystem that analyzes data but does not guide acquisition [8,11,13].
- **Advancement** — the Parameter Commons (O2) converts scattered, retirement-vulnerable expertise into versioned, DOI-citable records [15], raising everyday practice for expert and non-expert operators alike.
- **Promotion of microscopy** — open webinars, M&M tutorials, a Microscopy Today article, and a recommender whose every suggestion cites the primary microscopy literature carry MSA's name, and the case for rigorous microscopy, to students and to adjacent communities (additive manufacturing, materials, geoscience) that use these instruments daily but rarely engage with the Society (O4).

The "collaborative community" framing is taken literally: Commons records are contributed and reviewed by members, curated by a paid student corps, and governed by an MSA body from Year 2 onward (see Sustainability).

Alignment with the 2027 areas of emphasis

Call area	How this initiative addresses it
Area 2 — MSA-branded open source tools (primary)	CALIBER's RAG pipeline, parameter recommender, and benchmark tooling released under a permissive license as an MSA-branded resource for microscopy, microanalysis, data management, and image analysis
Area 6 — collection, preservation, dissemination of microscopy knowledge (primary)	The Parameter Commons captures technical procedures, basic theory, and the scientific background behind acquisition settings in versioned, DOI-citable records, cross-linked to MSA archived material (M&M proceedings, Microscopy Today)
Area 1 — student involvement	Paid student curation corps; curation is a structured on-ramp into quantitative microanalysis
Area 3 — member benefits	Members get a practical tool that shortens time-to-good-data on unfamiliar samples; contributor recognition for members
Area 5 — partnerships	Academic and national-lab partners contribute validation datasets and expert review [TODO: name partners]

Background and need

- **Parameters determine fidelity, quantitatively.** Standards-based SEM-EDS can reach about $\pm 5\%$ relative uncertainty; standardless can reach $\pm 30\%$, and normalized totals conceal errors [1]. EBSD throughput, frame averaging, voltage, binning, and above all phase-list selection each swing indexing success by tens of percentage points — one-frame patterns were almost entirely misclassified, and an over-inclusive 12-phase list produced zero correct assignments [2,3,4]. In

XRF of Al alloys, minor Mg has been over-reported by 40% relative and trace Mg by 5000%, while matrix-matched calibration holds major elements below 5% error [5,6,7].

- **The "why" is sparsely documented and hard to access.** Standards such as ISO 22309 and ASTM E1508 supply operating thresholds (e.g., overvoltage ≥ 1.8 , take-off near 35°) but not the derivations, uncertainty surfaces, or validation data needed to adapt them to a specific detector, alloy, and goal; investigators comparing two standardless systems could not even determine why their accuracies differed, because the algorithms and calibration databases were vendor-constrained [1]. Protocol repositories and methods sections preserve the procedural *what* but not the expert *why* — alternatives considered, failure modes, stopping criteria [8,11,14]. FAIR metadata schemas structure acquisition context but by design do not capture the causal rationale for choosing it [9,10]. And the practical expertise behind parameter choices demonstrably leaves science when experienced practitioners retire or move on [15].
- **Open tools stop short of guidance.** HyperSpy, kikuchipy, pyxem, EMsoft, DREAM.3D, ImageJ/Fiji, and napari are post-acquisition analyzers, simulators, or viewers; none recommends acquisition settings, grounds advice in citable literature, or quantifies its own uncertainty — and autonomous-microscopy work to date is task-specific and policy-driven rather than literature-grounded [8,11,13]. CALIBER is connective infrastructure over this ecosystem, not a duplicate of it.
- **The validation data don't exist yet.** No controlled, ground-truthed factorial study links acquisition settings to chemical and crystallographic fidelity for LPBF AlSi10Mg — published settings are examples, not validated optima [1,3,4]. The open benchmark in this initiative creates exactly that dataset, for the community.
- **Much of the source knowledge is already MSA's.** The strongest parameter studies and ecosystem papers appear in *Microscopy and Microanalysis* and M&M proceedings [3,12,13] — MSA archived material that the Commons would organize, cross-link, and make actionable (Area 6).

Objectives and deliverables

1. **O1 — MSA-branded open-source CALIBER release.** Public repository (permissive license) containing the retrieval-augmented recommender, uncertainty-aware feedback loop, and corpus-building tools; MSA branding and hosting arrangement agreed with Council; built to interoperate with the existing open ecosystem (HyperSpy/kikuchipy data models, NeXus/NXem metadata [9]) rather than replace it. *Deliverables:* v0.1 (Y1Q2), v1.0 (Y2Q2); *documentation and install-free web demo.*
2. **O2 — The Microanalysis Parameter Commons.** A public, versioned knowledge base of parameter-rationale records (per modality $\times$ parameter $\times$ material class: the recommended range, the physics behind it, the primary sources, and known failure modes), with DOIs per release, a contribution/review workflow, and cross-links into MSA archives (M&M proceedings, Microscopy Today). *Deliverables:* *schema + 50 seed records* (Y1Q2), *150+ records spanning SEM, EDS, EBSD, XRF* (Y2Q4).

3. **O3 — Open validation benchmark.** A published benchmark on novel alloy samples (laser powder bed fusion Al-Si and the broader alloy campaign) that fills the documented absence of controlled factorial acquisition studies [1,3]: raw spectra/patterns archived and re-analyzed offline outside vendor software, so recommended vs. naive parameters can be compared reproducibly; early-stopping quality heuristics (e.g., EBSD confidence-index screening after partial scans) evaluated and reported. *Deliverables: open dataset with FAIR metadata per MaRDA recommendations [9] (Y2Q1); benchmark report.*
4. **O4 — Dissemination, training, and promotion of microscopy.** M&M tutorial/workshop, a Microscopy Today article, and a student curation corps designed around what makes trainee curation sustainable: bounded tasks, structured mentorship and review, version control, and visible credit on DOI'd releases [15,16,17]. Two open webinars — run through MSA's online channels and open to non-members — and course-ready tutorial notebooks point the resource outward, toward students and adjacent communities (additive manufacturing, materials, geoscience) that depend on SEM/EDS/EBSD/XRF but rarely interact with MSA; because every recommendation cites the literature behind it, each use of the tool is itself an introduction to microscopy scholarship. *Deliverables: workshop at M&M [2027 and/or 2028] ; two open webinars (Y1Q4, Y2Q3); tutorial notebooks; 6–10 student curators trained.*

Work plan and timeline (two years)

Period	Milestones
Y1 Q1	MSA branding/hosting agreement; Commons record schema; seed extraction for the highest-impact parameters (EDS: accelerating voltage + [TODO: second parameter under evaluation]; EBSD: beam current, voltage, phase list; SEM imaging baseline set)
Y1 Q2	CALIBER v0.1 open-source release; Commons alpha with 50 seed records; student curator recruitment
Y1 Q3	Validation campaign 1 (LPBF Al-Si): archived raw data, offline re-analysis, uncertainty-aware feedback loop on recommended settings
Y1 Q4	Public beta; M&M tutorial submission; Microscopy Today article draft; first open webinar
Y2 Q1	Benchmark dataset release (FAIR metadata, DOI); community contribution workflow opens
Y2 Q2	CALIBER v1.0; XRF modality coverage; 100+ records
Y2 Q3	M&M workshop; second open webinar; assessment against success metrics
Y2 Q4	150+ records; sustainability handoff to [TODO: MSA committee/FIG] ; final report to Council

Budget (draft — \$20,000 over two years)

Item	Amount
Student curation and development micro-stipends (6–10 curators × 2 yr)	\$8,000
Instrument time for the open validation benchmark (SEM/EDS/EBSD/XRF sessions)	\$6,000

Item	Amount
Hosting, compute, and DOI/archiving costs (2 yr)	\$2,000
M&M tutorial/workshop materials and costs	\$2,500
Dissemination (open-access fees, Microscopy Today piece, webinar production)	\$1,500
Total	\$20,000

[TODO: adjust to institutional rates; confirm no overhead applies; detailed justification per MSA form]

Benefit to MSA and success metrics

- Positions MSA as the home of *preserved, citable* microanalysis practice — knowledge that currently evaporates with retirements [15] — under its own brand (Areas 2 and 6), following the model of societies that built trusted open resources (rOpenSci, MolSSI, ELIXIR) [16,18].
- Carries MSA's brand — and the case for rigorous, literature-grounded microscopy — into adjacent research communities that run these instruments daily without engaging the Society; the open webinars and the tool's cite-the-source design make this a standing visibility and recruitment channel, directly serving the mission's promotion of microscopy.
- A concrete, recurring member benefit: shorter time-to-reliable-data on unfamiliar samples.
- Metrics: number of published Commons records (target 150+); tool users/downloads and web-demo sessions; benchmark dataset downloads; workshop and webinar attendance, including the non-member share reached; student curators trained; records cross-linked to MSA archived material; new/retained members reporting use [TODO: baseline survey mechanism].

Sustainability

Year 2 transitions governance to an MSA body

[TODO: identify committee or Focused Interest Group] with a lightweight editor-plus-reviewers model. The precedent literature is specific about what makes society-branded resources survive — governance by a trusted organization, integration with existing community activities (here: M&M, Microscopy Today), contributor recognition, and maintenance capacity — and about the failure mode to avoid: volunteer-only efforts detached from established infrastructure [16,18]. Post-award hosting costs are minimal (static site + repository); curation continues through the student pipeline and member contributions credited on DOI'd releases.

Team, partners, and reviewers

[TODO: PI and co-applicants; partner labs and agencies (Area 5) – candidates from the byu-vcl shortlist; note the call excludes microscopy vendors as partners]

References

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Appendix: supporting research artifacts

Full Edison reports backing this draft, in [outputs/msa_2027_strategic_initiatives/](#):

- [need_evidence_answer.md](#) — quantitative evidence report with a proposal-ready needs statement; includes the evidence table ([need_evidence_artifact-00.md](#))
- [deep_landscape_answer.md](#) — landscape review with ten distilled gap statements ([deep_landscape_artifact-00.md](#)) and a tool-by-tool gap table ([deep_landscape_artifact-01.md](#))